

COLLEGE / VLG PROJECT 2025-26

# How Trendora was built

Step-by-step construction log for the viva.

React 18 - Vite 5 - Context API - four login authorities.

Department of Computer Science / Information Technology

## Contents

1. Certificate
2. 1. Introduction
3. 4. System design
4. 5. Implementation
5. Appendix A - How to run
6. Appendix B - User manuals by role

## Certificate

This is to certify that the project titled "Trendora - A Multi-Authority Online Shopping Website" is a bona fide record of work carried out as part of the academic curriculum. The software implements a customer storefront inspired by Myntra, Meesho and Flipkart together with Seller, Admin and Owner consoles. The candidate has presented the working system, test cases and this report for evaluation.

# 1. Introduction

## 1.1 Motivation

College students already shop on Meesho, Myntra and Flipkart. Cloning only the storefront leaves the authority model unexplained. Real platforms separate the shopper, the seller, the operations admin and the owner. Trendora therefore implements all four logins.

## 1.2 Problem statement

Design a responsive multi-role shopping website that covers the shopping journey of Indian fashion/general e-commerce and the back-office of a marketplace, without a live warehouse or payment gateway.

## 1.3 Objectives

1. Study IA of Myntra, Meesho and Flipkart.
2. Implement customer features: catalogue, search, filter, sort, PDP, size chart, PIN check, reviews, Q&A, wishlist, compare, bag, coupons, gift wrap, checkout, orders, cancel, return, addresses, Insider, studio, offers, help.
3. Implement seller studio: list SKUs, own orders, earnings after commission.
4. Implement admin desk: catalogue, order pipeline, returns, coupons, tickets, user block.
5. Implement owner console: all admin rights plus create staff, change roles, store settings, GMV reports.
6. Persist session data locally and ship PDF / Word / PPT documents.

## 1.4 Scope

In scope: the modules above, INR pricing, simulated payments, college documents.

Out of scope: real Razorpay capture, AWB logistics, native apps, ML recommendations.

## 4. System design

### 4.1 Architecture

Browser - React pages - StoreContext (use cases) - products.js + localStorage.

Three layers: presentation, application state, persistence.

### 4.2 Routing

Public: /, /shop, /product/:id, /offers, /brands, /studio, /insider, /compare, /help, /login, /register.

Customer: /cart, /checkout, /orders, /returns, /addresses, /wishlist, /profile, /notifications.

Guarded: /seller/\*, /admin/\*, /owner/\* via RequireRole.

### 4.3 Price algorithm

$grand = \max(0, subtotal - coupon + shipping + giftWrap)$

$shipping = 0$  if  $subtotal \geq freeShipMin$  else  $shipFee$  (owner configurable; default 999 / 79).

$giftWrap = 49$  if selected.

$Insider\ points += \text{floor}(grand / 10)$ .

$Seller\ net = item\ sale - commission\%$  (default 12).

### 4.4 Order state machine

Confirmed - Packed - Shipped - Out for delivery - Delivered - (optional) Returned.

Cancel allowed before Shipped. Return allowed only after Delivered.

### 4.5 Demo accounts

demo@trendora.in / demo123 (Customer)

seller@trendora.in / seller123 (Seller)

admin@trendora.in / admin123 (Admin)

owner@trendora.in / owner123 (Owner)

## 5. Implementation

Environment: Node.js 18+, npm, Vite 5, React 18, React Router 6, Context API, CSS design tokens.

Key modules:

src/context/StoreContext.jsx - every mutation (auth, bag, orders, catalogue, tickets).

src/data/products.js - seed catalogue, coupons, size charts, PIN helper.

src/pages/dash/\* - Owner / Admin / Seller consoles.

src/components/RequireRole.jsx - route guard.

Security honesty: demo passwords sit in localStorage. Production would hash on a server (bcrypt) and issue JWT/cookies. This is stated in FAQ, slides and viva notes.

## Appendix A - How to run

cd new-shopping-website

npm install

npm run dev

Open the printed URL (default <http://localhost:5173>).

Production: npm run build && npm run preview.

## Appendix B - User manuals by role

Customer: Login demo@trendora.in - shop - PDP - bag - FESTIVE20 - checkout - track.

Seller: Login seller@trendora.in - Catalogue - list SKU - Earnings.

Admin: Login admin@trendora.in - Orders - mark Packed / Shipped / Delivered - Returns / Tickets.

Owner: Login owner@trendora.in - People & roles - create staff - Settings - Reports.